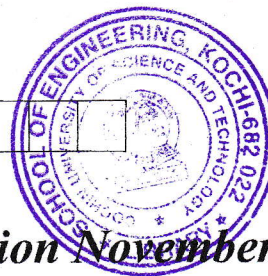


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C

B.Tech. Degree III Semester Examination November 2018

ME 15-1305 METALLURGY AND MATERIAL SCIENCE (2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Calculate the density of BCC iron if its atomic radius is 0.124 nm.
- (b) Sketch the following directions and planes in a cubic crystal.
 $(2\ 2\ 1)$, $(1\ 0\ 1)$, $[2\ 3\ 4]$, $[1\ 0\ 0]$
- (c) Define the term phase and explain Gibb's phase rule.
- (d) What are dislocations? How are they classified? Explain their role in plastic deformation of metals.
- (e) State and explain Fick's laws of diffusion.
- (f) Explain Hume-Rothery rules.
- (g) Differentiate between cold working and hot working.
- (h) What do you mean by ductile-brittle transition?
- (i) Explain different types of cast irons.
- (j) What are ceramics? State the important properties of ceramics.

PART B

(4 × 10 = 40)

- II. (a) Differentiate between homogeneous and heterogeneous nucleation.
 - (b) Explain the mechanism of diffusion.
- OR**
- III. (a) Discuss the atomic packing factor and co-ordination number of BCC and FCC crystal structures.
 - (b) Discuss the factors affecting diffusion.
- IV. (a) Draw Fe-C phase diagram and state important reactions.
 - (b) Explain austempering and martempering.
- OR**
- V. (a) Draw the equilibrium diagram of a binary alloy with complete solid solubility and explain.
 - (b) Differentiate between annealing and normalizing.
- VI. (a) Discuss elastic, plastic and viscoelastic behaviour of metals.
 - (b) Draw a creep curve and explain. Name two creep resistant materials.
- OR**
- VII. (a) Discuss three strengthening mechanisms.
 - (b) Explain recovery, recrystallisation and grain growth.
- VIII. (a) What are the different types of composites? Give one application for each type.
 - (b) Write short notes on:
 - (i) Smart materials
 - (ii) Biomaterials
- OR**
- IX. (a) What is stainless steel? Explain the different types of stainless steels.
 - (b) Write short notes on:
 - (i) Aluminium alloys
 - (ii) Super alloys